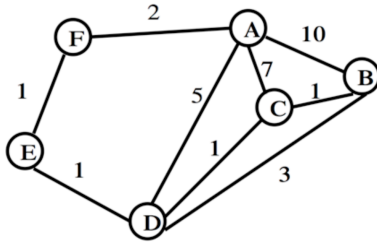


ECE-UY 3613, Spring 2024, Quiz 4

Name:

ID:

1. Apply Dijkstra's shortest path routing algorithm to the following network to construct the shortest paths from node A to all the other nodes in the network.
 - a) show the steps of Dijkstra algorithm's operation by providing a calculation table similar to that in the lecture;
 - b) show the forwarding table on node A;
 - c) plot the shortest path tree from A to all the other nodes.



(a) Traversed	L_{AA}, P_{AA}	L_{AB}, P_{AB}	L_{AC}, P_{AC}	L_{AD}, P_{AD}	L_{AE}, P_{AE}	L_{AF}, P_{AF}
A	0	10, A	7, A	5, A	∞	(2, A)
A, F	0	10, A	7, A	5, A	(3, F)	
A, E, F	0	10, A	7, A	(4, E)		
A, D, E, F	0	7, D	(5, D)			
A, D, E, F,	0	(6, C)				
A, B, C, D, E, F						

(b)	Destination	Node	Next-hop
	B		A
	C		F
	D		F
	E		F
	F		F

(c)

```

graph LR
    A((A)) ---|2| F((F))
    F ---|1| E((E))
    E ---|1| D((D))
    D ---|1| C((C))
    C ---|1| B((B))
  
```